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Thermal Structure of Saturn from Voyager Infrared Measurements: Implications for Atmospheric Dynamics

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Saturn atmospheric temperatures at the 150-mbar level retrieved from Voyager IRIS measurements indicate the presence of small-scale meridional gradients which are approximately symmetric with respect to the equator, but are superposed on a large-scale hemispheric thermal asymmetry. Under the assumption that the retrieved values at this atmospheric level represent kinetic temperatures on a constant pressure surface, it is suggested that the small-scale structure is produced by a meridional circulation associated with the dissipative decay of the zonal winds with height, while the hemispheric asymmetry represents a thermal response to the seasonally varying insolation. The small-scale gradients are correlated with zonal winds derived from Voyager images at mid and high latitudes through the thermal wind relation; the calculated thermal wind shears suggest a decay with height of the jet system toward a state of uniform eastward flow. The existence of the approximately symmetric zonal winds and associated temperature gradients in the presence of a large-scale seasonal thermal response suggests that the jet system is driven at depths substantially below the levels where seasonally modulated insolation is important ($p \gtrsim 0.5$ bar).

INTRODUCTION

Atmospheric temperatures retrieved from Voyager infrared spectrometer (IRIS) measurements were used to establish a correlation between thermal structure and zonal winds for both hemispheres of Jupiter and a portion of the northern hemisphere of Saturn (Pirraglia *et al.*, 1981). Upper tropospheric meridional temperature gradients are found to be related to the systems of zonal jets on both planets through the thermal wind equation. Unlike Jupiter, Saturn possesses an apparent large-scale thermal asymmetry in the upper troposphere with the northern hemisphere colder than the southern hemisphere, a result which is qualitatively consistent with Saturn's obliquity, which is large (27°) compared to that of Jupiter (3°) (Hanel *et al.*, 1981, 1982). Even though a large-scale thermal asymmetry may exist, the zonal wind system derived from cloud motions observed with Voyager images (Smith *et al.*, 1982) appears to be highly symmetric with respect to the equator. It is important to establish whether or not the correlation between the smaller-scale temperature gradients and the northern hemisphere jets also exists in the southern hemisphere of Saturn. Data employed in a preliminary analysis of the southern hemisphere (Hanel *et al.*, 1981) did not possess adequate spatial resolution to yield an unambiguous answer to this question. Recent analyses which combine Voyager 1 and 2 data permit the study of thermal structure at higher spatial resolution over portions of the southern hemisphere and also extend the coverage to high northern latitudes. It is the objective of this report to present the results of these analyses and to discuss their possible implications for the upper tropospheric meteorology of Saturn and its relation to the dynamics of the deeper atmosphere.

ANALYSIS OF THERMAL STRUCTURE

Atmospheric temperature profiles were obtained by inversion of radiances from

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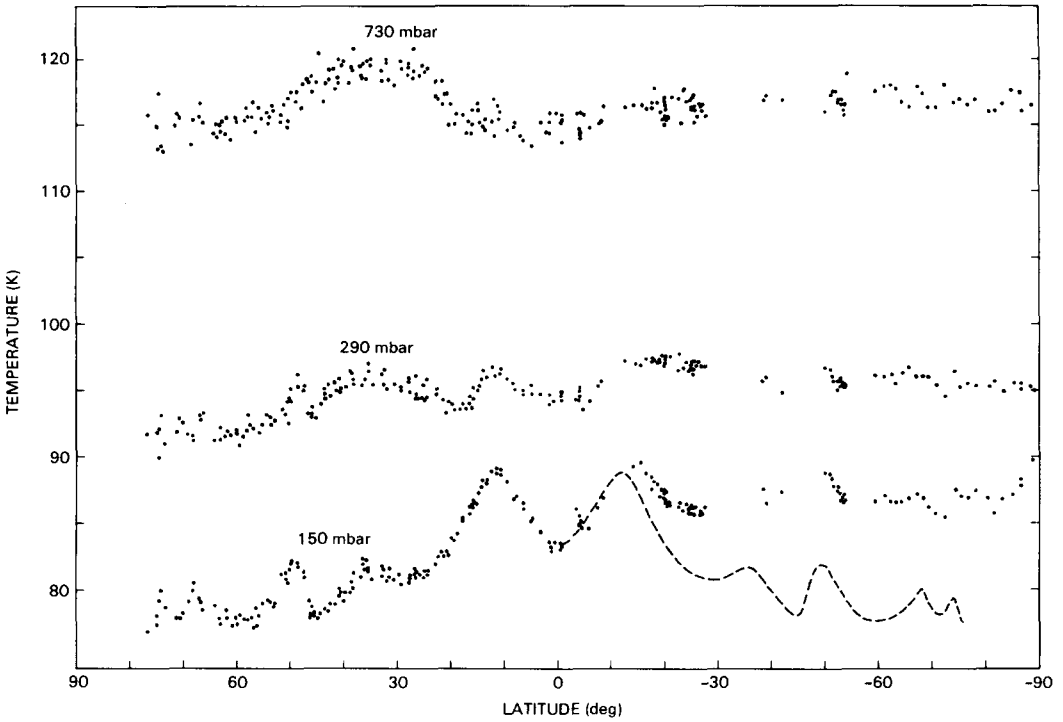


FIG. 1. Temperatures retrieved by the inversion of Voyager infrared spectral measurements. The values shown represent mean values for atmospheric layers approximately one-half scale height thick. The broken curve is a fit to the northern hemisphere 150-mbar temperatures which has been folded over at the equator to permit a comparison with southern hemisphere results.

within selected regions of the IRIS spectra, using techniques similar to those discussed by Conrath and Gautier (1980). Tropospheric temperatures are derived from measurements within the 200- to 600-cm^{-1} spectral region, where the atmospheric opacity is assumed to be entirely due to collision-induced hydrogen transitions. The effective vertical resolution of the retrievals used in this study is about half of the pressure scale height, and the formal errors due to instrument noise propagation are less than 1°K for inversions of individual spectra.

The latitudinal dependence of the retrieved temperatures is displayed in Fig. 1 for selected pressure levels. The data include measurements from both Voyagers 1 and 2 and were selected to have a spatial resolution of approximately 4° of latitude or better. Coverage at this spatial resolution in the southern hemisphere is considerably less complete than in the north, due in part to loss of Voyager 2 data resulting from a scan platform malfunction.

For pressures less than about 300 mbar, there is a large-scale equator-to-pole decrease in temperature in the northern hemisphere while essentially no equator-to-pole gradient is observed in the south. Despite this large-scale hemispheric asymmetry, the data suggest that the superposed smaller-scale structure is approximately symmetric with respect to the equator. In Fig. 1 the broken curve represents a fit to the northern hemisphere 150-mbar temperatures which has been folded over at the equator for comparison with the data in the southern hemisphere. Temperature gradients appear to occur at approximately the same latitudes in both hemispheres, at least in those regions where data coverage permits a comparison.

DISCUSSION

Possible contributions to the atmospheric opacity by clouds and hazes have been neglected in retrieving the temperatures shown in Fig. 1. The viewing geometries for the data taken in the northern and southern hemispheres were similar so systematic differences in slant paths cannot be responsible in any way for the observed hemispheric differences. However, if haze opacities are important and are horizontally inhomogeneous, the inference of spurious latitudinal temperature structure can result. Comparisons of IRIS-derived temperatures with results from the Voyager radio occultation experiment are potentially useful in assessing aerosol opacity effects. The Voyager 2 ingress and egress radio occultation temperature profiles at 36°N and 31°S, respectively, are nearly identical in the troposphere (Tyler *et al.*, 1982). While temperatures derived from IRIS measurements are essentially in agreement with the occultation profile at 36°N, at 31°S the IRIS results are slightly cooler than the occultation values at 730 mbar, and significantly warmer in the upper troposphere. Enhanced tropospheric haze or clouds in the south could result in the underestimation of temperatures at the deeper levels, but cannot account for warmer inferred temperatures in the upper troposphere. The latter could possibly result from the presence of a stratospheric infrared absorber with an appropriate spectral dependence as suggested by Orton (1983). However, the analyses of West *et al.* (1982a,b), using Voyager and Pioneer photopolarimeter results and ground-based spectrophotometry in near-infrared methane bands, suggest the absence of stratospheric aerosols in this region. In fact the results appear to indicate that the stratosphere and upper troposphere are more nearly free of particulates at mid-southern latitude than in any other region on the planet.

For purposes of this discussion we shall assume that the retrieved 150-mbar temperatures in Fig. 1 represent kinetic temperatures on a constant pressure surface. However, the resulting conclusions must be regarded as tentative until a better understanding of the particulate contributions to the infrared opacity can be achieved.

The seasonal response implied by the large-scale asymmetry in the 150-mbar temperatures can be understood through the consideration of radiative relaxation times. Within the framework of the linearized Newtonian cooling approximation to the atmospheric radiative transfer (Goody, 1964), the thermal response to the component of insolation varying with the seasonal frequency ω_S is dependent on the parameter $\omega_S\tau_R$, where τ_R is the radiative relaxation time. The resulting annual temperature wave lags the solar forcing by a phase shift $\delta = \tan^{-1} \omega_S\tau_R$, and the amplitude is inversely proportional to $\omega_S\tau_R$ for large $\omega_S\tau_R$. Carlson *et al.* (1980) have calculated the phase-lagged stratospheric thermal structure of Saturn appropriate for the seasonal conditions at the time of the Voyager 1 encounter. In the present study we are interested in the upper troposphere so we have used the calculations of Gierasch and Goody (1969) for hydrogen atmospheres with a scaling appropriate to Saturn. The parameter $\omega_S\tau_R$ and the resulting phase lag are shown as functions of atmospheric pressure level in Fig. 2. Examination of the figure indicates that in the upper troposphere an observable thermal response is expected with a phase lag of about one-sixth of a Saturnian year or 5 terrestrial years. At levels deeper than about 300 mbar, however, the amplitude of the response should be greatly reduced. Since the subsolar point crossed the Saturnian equator moving into the northern hemisphere approximately 8 months before the Voyager 1 encounter and 17 months before that of Voyager 2, the observed relatively cold northern upper troposphere is qualitatively consistent with this picture. To determine the detailed latitudinal dependence of the thermal response, a more nearly com-

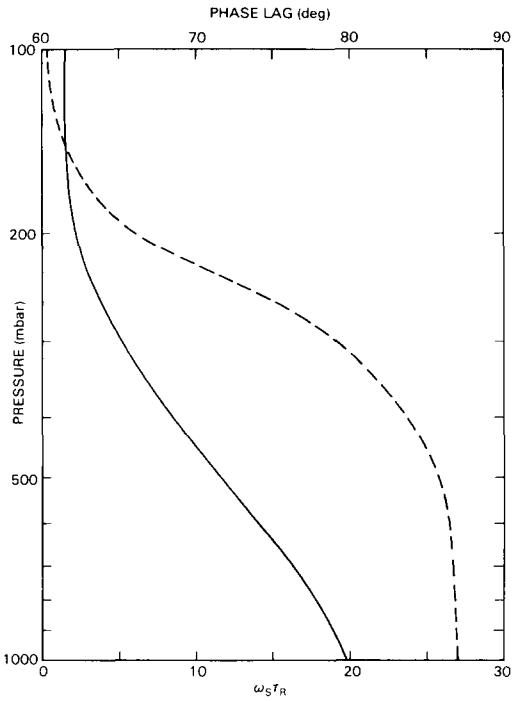


FIG. 2. Parameters related to the rdiative response of the upper troposphere of Saturn to forcing by the seasonally varying insolation. The solid curve shows the variation with atmospheric pressure level of $\omega_S \tau_R$, where ω_S is 2π/(Saturnian year) and τ_R is the radiative time constant. The broken curve represents the phase lag of the thermal response where 90° corresponds to one-fourth of the Saturnian year.

plete radiative transfer calculation would be required in which the ring-modulated latitudinal dependence of the insolation is taken into account. Such a study is outside the scope of this paper.

Under the assumptions of geostrophic balance and hydrostatic equilibrium, the eastward wind u is related to the latitudinal temperature gradient along a constant pressure surface through the thermal wind relation

$$\frac{\partial u}{\partial z} = -\frac{g}{afT}\frac{\partial T}{\partial \lambda}, \qquad (1)$$

where z is height, g is gravitational acceleration, a is planetary radius, f is the Coriolis frequency, T is temperature, and λ is latitude. Temperature gradients estimated from the 150-mbar temperatures of Fig. 1 yield the thermal wind shears shown in Fig. 3. This represents an extension to the southern hemisphere as well as to higher northern latitudes of the previous results of Pirraglia *et al.* (1981). On the right-hand side of Fig. 3, eastward wind velocities obtained from observations of cloud top motions using Voyager images (Smith *et al.*, 1982) are shown for comparison. At latitudes north of about 30° the pattern of the thermal wind shears correlates with that of the jet system to within the uncertainties introduced by inaccuracies in data location. To the extent that the data coverage permits a comparison, a correlation also appears to exist in the southern hemisphere. With the exception of the regions between about 10 and 25° latitude in both hemispheres, the sense of the shear is such as to reduce the relative amplitudes of the eastward and westward jets with increasing height. Note that at latitudes above ~30°, both positive and negative values of the vertical shear occur at approximately the same magnitude, while the winds themselves are strongly biased toward eastward flow. Therefore, if the shears persist over ~6 scale heights, the jets will decay not to

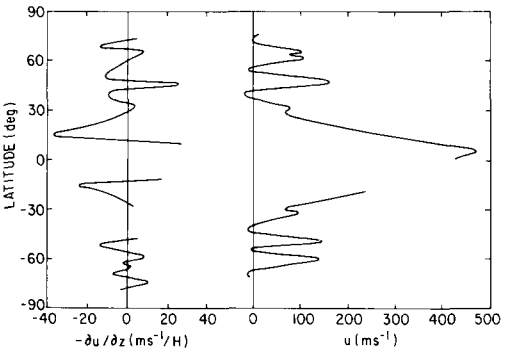


FIG. 3. Vertical thermal wind shear at the 150-mbar level (left) compared with zonal wind speed at the cloud tops (right). The thermal wind shear was calculated from the temperatures shown in Fig. 1, and the zonal winds were obtained from observations of cloud motion (Smith *et al.*, 1982). Note that the negative of the shear has been plotted for this comparison.

zero velocity but to a finite eastward motion in a reference frame defined by the radio rotation period of the planet.

The correlation between the temperature structure and the observed zonal wind suggests that the structure must result either directly or indirectly from dynamical processes. One possibility is that the observed upper tropospheric thermal gradients are dynamically forced by deeper atmospheric layers. As an example we shall consider the thermal response of a passive, statically stable, dissipative upper atmosphere forced by an imposed zonal jet at its lower boundary. In general the response will depend on the nature of the dissipative processes operative in this portion of the atmosphere. The apparent tendency of the latitudinal gradients of the zonal wind to decrease with height suggests that horizontal momentum diffusion may be important. We shall assume that this process can be represented by a linear drag or Rayleigh friction term which, for quasigeostrophic flow, must be balanced to lowest order by the Coriolis acceleration acting on a meridional motion, i.e.,

$$fv = u/\tau_M, \tag{2}$$

where v is northward meridional velocity and τ_M is a Rayleigh friction damping time. Through requirements of mass continuity, there will be a vertical motion w associated with the frictionally controlled meridional velocity as shown schematically in Fig. 4. Downward and upward motion will result in adiabatic heating and cooling, respectively. This heating and cooling must in turn be balanced by radiative damping in the absence of diabatic heat sources and sinks. If the radiative damping is parameterized as Newtonian cooling, the lowest-order balance can be written

$$w\left(\frac{dT_0}{dz} + \frac{g}{c_p}\right) = -\frac{\Delta T}{\tau_R}, \tag{3}$$

where T_0 is the horizontally averaged temperature, w is vertical velocity, c_p is specific heat at constant pressure, and ΔT is the temperature perturbation due to atmospheric motion. For example, an eastward jet must be accompanied by a northward meridional velocity according to (2), which must in turn be accompanied by upward motion to the south of the jet and descending motion on the north (see Fig. 4). From (3) this yields relatively low temperatures south of the jet with higher temperatures to the north. The resulting temperature gradient, according to the thermal wind equation (1), leads to a reduction in the zonal wind speed with height. A similar dynamical scaling has been examined in detail by Conrath *et al.* (1981) with applications to the behavior of large-scale features in the Jovian atmosphere.

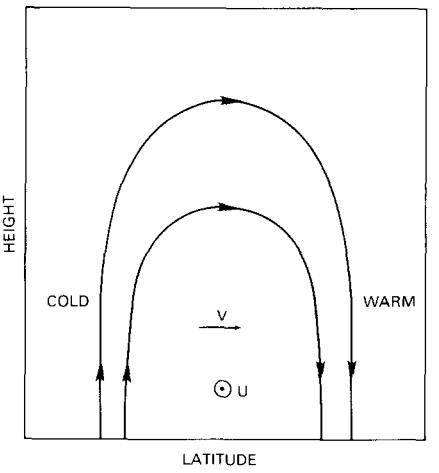


FIG. 4. Schematic of the meridional circulation associated with the decay with height of zonal jet. The zonal wind u is directed out of the plane of the diagram and the northward meridional wind is denoted by v.

If the above scenario is correct, the observed small-scale temperature gradients in the upper troposphere represent a response to dynamic forcing by a deep system of symmetric zonal jets. This jet system is subject to seasonally varying modulations in the upper atmosphere, but these will be small relative to the total energy of the zonal flow, permitting the jets to retain their essentially symmetric character. The actual

depth of the circulation remains an open question as far as this analysis is concerned. The results would apparently be consistent with proposed deep cylindrical circulation models (Busse 1976; Ingersoll and Pollard, 1982), but other models cannot be ruled out as long as the winds are driven at levels substantially deeper ($p \gtrsim 0.5$ bar) than the levels affected by the seasonally varying component of insolation.

Vertical motion associated with the jet system may play a significant role in the distribution of clouds and condensation particulates. It is interesting to note that there is a strong correlation between the smaller-scale temperature structure in the upper troposphere and absorption optical depths at 2640 Å inferred from Voyager 2 photopolarimeter data in the northern hemisphere (West *et al.*, 1982b). Relatively small optical depths at a given pressure level correspond to relatively high temperatures and presumably descending motion, while larger optical depths correspond to lower temperatures and rising motion. Such a behavior might be expected if the absorber is a condensation product transported vertically from the condensation level.

At levels deeper than 300–400 mbar, the temperature gradients no longer correlate with the zonal jets. As can be seen in Fig. 1, large-scale gradients appear to exist which are not symmetric with respect to the equator. It is likely that these are due, at least in part, to contributions to the infrared opacity by hazes and clouds which were not taken into account in the temperature retrievals. The large-scale latitudinal temperature structure at 730 mbar is correlated with the large-scale envelope of the absorption optical depths for the northern hemisphere derived from the Voyager 2 photopolarimeter (Lane *et al.*, 1982; West *et al.*, 1982b). The largest optical depths are found at high latitudes and near the equator, coincident with relatively cool temperatures. In addition, ground-based spectrophotometry in near-infrared methane bands indicates the presence of relatively high altitude aerosols or clouds in the equatorial regions (West *et al.*, 1982a).

CONCLUSIONS

We now summarize the principal conclusions of this study. Meridional temperature gradients which are correlated with the zonal jets through the thermal wind equation are essentially symmetric with respect to the equator. Calculated thermal wind shears imply a decrease in the latitudinal gradients of the zonal wind with height. We suggest that the thermal structure represents a response to forcing by a deep-seated dynamical regime which is responsible for the symmetric zonal jet system. A superposed large-scale asymmetry in the thermal structure of the upper troposphere is qualitatively consistent with a phase-shifted response to the seasonally varying insolation. The approximate symmetry of the zonal jets and associated temperature gradients in the presence of a seasonal thermal response implies that the zonal wind system is driven at levels which are sufficiently deep to be unaffected by the seasonally varying insolation. The above conclusions are based on the assumption that the retrieved temperatures near 150 mbar are not seriously influenced by the presence of atmospheric aerosols. At pressures greater than about 300 mbar, the retrieved thermal structure is not simply correlated with either the jets or the solar input and may be influenced by contributions to the infrared opacities from clouds.

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